



CSE471: System Analysis and Design
Assignment on Functional Requirements
Proposed Project Title: Job Sensei

Group No: 8, CSE471 Lab Section: 02, [Summer] [2026]		
SL	ID	Name
1	22301729	Taznia Tabassum Onon
2	22301675	Nazifa Anjum Prova
3	22301190	Adreed Saadad Hasan

Submission Date: 5th July, 2026

Project Overview

"Job Sensei" is a job-hunting and career-support application that helps users search for suitable job opportunities, create and optimize resumes, identify skill gaps, receive personalized learning resource recommendations, and track job applications. The platform also provides an AI-powered conversational career companion for career guidance and a skill-based community where users can share knowledge, discuss career-related topics, and learn from one another. In addition, the system imports job listings from public job APIs, calculates job-match scores, and helps users improve their employability through continuous learning and career development. The application is built using Flutter for the front end, Node.js with Express.js for the back end, MongoDB Atlas as the database, and integrates external APIs such as the Gemini API, Remote Jobs API/Arbeitnow Job Board API, YouTube Data API, and Gmail API/SMTP services where appropriate.

Tech Stack:

- Language: Dart
- Framework: Flutter for front-end, Node.js with Express.js for back-end
- Styling: Flutter Material Design components
- Database: MongoDB Atlas
- ORM: Prisma
- Deployment: Android APK(local) + Backend (vercel)
- External APIs:
 - **Remote Jobs API / Arbeitnow Job Board API:** Free public APIs for importing job listings and job requirement data.
 - **Gemini API:** For an AI-powered conversational career companion that provides career guidance and answers user queries
 - **YouTube Data API:** For recommending relevant free learning videos based on missing skills.
 - **Gmail API / SMTP Service:** For account emails, job alerts, and application status notifications.

User Roles

1. **Student / Job Seeker:** The primary users - they can create profiles, search for jobs, build resumes, apply to jobs, track applications, learn missing skills, join communities, and use the AI companion.

2. **Recruiter / Employer:** A verified job provider - they can create organization profiles, post job openings, review applicant resumes, shortlist candidates, and update application statuses.
3. **Admin:** A system super-user - responsible for approving employers, monitoring job posts and communities, managing reported content, and maintaining learning-resource categories.
4. **Community Member :** Any job seeker or user also acts as a community member by joining skill-based groups, sharing useful resources, posting questions, answering discussions, and learning together with other users.

Functional Requirements

SL	Common Workflows
1	User Authentication:Users register a new account, log in securely, recover forgotten passwords if needed, and access the system according to their assigned role.
2	Profile Management:After logging in, users complete and update their profiles. Job seekers manage their career information, while recruiters maintain their company profiles. The system stores this information for personalized services and platform features.
3	Admin Verification and Access Management: Administrators review recruiter registration requests, verify organization details, monitor platform activities, and manage user permissions to ensure the security and reliability of the system.

Module 1	
Member	Feature Description
Adreed Saadad Hasan ID:22301190	The system stores a complete career profile for each job seeker, including education, work experience, preferred job roles, location, expected salary, skills, certifications, portfolio links, and career goals. Users edit this profile whenever their background changes, and the profile becomes the base data for job matching, resume suggestions, and skill-gap analysis.
Taznia Tabassum Onon ID: 22301729	The system integrates the Gemini API to provide an AI-powered conversational assistant that interacts with users through a chat interface. It processes user queries, generates context-aware responses, and facilitates natural language interaction with the system. The integration is modular, allowing future AI-powered functionalities to be incorporated without significant architectural changes.
Nazifa Anjum Prova ID: 22301675	The system provides a resume builder where users create, edit, preview, duplicate, and save multiple resume versions using information from their career profile. Users can customize each resume for different career fields before applying for jobs.

Module 2	
Member	Feature Description
Adreed Saadad Hasan ID:22301190	The system imports job listings from public job APIs such as Remote Jobs API or Arbeitnow Job Board API. It stores important job details such as title, company, location, job type, remote status, required skills, responsibilities, deadline, and source link so users see both internal and external opportunities in one place.
Taznia Tabassum Onon ID: 22301729	The system provides skill-based community groups where users can join discussions based on technologies, job roles, or career interests. Users can create posts, ask questions, answer discussions, share learning resources, and exchange career experiences with other members. The system also allows users to search communities and follow discussions.
Nazifa Anjum Prova	The system uses an AI API to analyze a selected job description and compare it with the user resume. It suggests stronger keywords, relevant

ID: 22301675	project highlights, skill ordering, and short summary improvements so the resume fits the selected job more closely.
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Module 3	
Member	Feature Description
Adreed Saadad Hasan ID:22301190	The system provides a searchable job list where users filter opportunities by title, company, skill, location, salary range, job type, experience level, and remote or on-site preference. Search results show the most relevant job details first and allow users to save interesting jobs for later review.
Taznia Tabassum Onon ID:22301729	The system analyzes the user's existing skills against the requirements of a selected job or target career path to identify missing competencies. It categorizes missing skills into technical skills, software tools, certifications, and soft skills, prioritizes them according to their importance for the selected role, and explains how each missing skill affects the user's overall suitability.
Nazifa Anjum Prova ID: 22301675	The system lets users save recommended resources into a personal learning plan and mark each resource as not started, in progress, or completed. Progress tracking connects completed resources with target job skills, updates the user learning history, and supports reward points for consistent improvement.

Module 4	
Member	Feature Description
Adreed Saadad Hasan ID:22301190	The system compares the user's career profile and selected resume with the requirements of a selected job and generates an overall match score. It highlights the user's strongest qualifications and indicates the overall suitability for that job without providing learning recommendations.

Taznia Tabassum Onon ID:22301729	The system recommends personalized learning resources based on the missing skills identified during Skill Gap Analysis. It retrieves relevant tutorials from the YouTube Data API and organizes each recommendation with the related skill, resource title, platform, estimated difficulty level, and direct learning link, helping users prepare for their target jobs.
Nazifa Anjum Prova ID: 22301675	The system tracks each submitted application through statuses such as submitted, under review, shortlisted, rejected, interview scheduled, and selected. Users receive free email notifications through Gmail/SMTP or EmailJS when recruiters update the status or when an interview reminder becomes available.